

Class Test

Image Processing & Computer Vision

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- Q1) Frequency domain filtering :- It is a type of filtering method in which we modify a signal ~~after~~ after transforming it into frequency domain. The types of frequency domain filters are
- (1) Low pass filter (LPF)
 - (2) High pass filter (HPF)

Applying low pass filter and high pass filter effects an image as:-

When we apply low pass filter in the image it removes the blurring of image and smoothens the image.

Whereas,

When we apply high pass filter in the image we can detect the edges of the image and sharpen the image.

- Q2) Step by step procedure to perform filtering in frequency domain using FFT & inverse FFT are as follows:-

FFT $\rightarrow$ It is also known as fast Fourier transform that converts a signal from its original domain into frequency domain.

For eg:- If we take a sin wave noise = $20 * \sin(\text{sp. linspace}(0, 50, 102))$ and apply FFT to it, the output will not be a wave instead

Page _____

Q2) Step by step procedure to perform filtering in frequency domain using FFT & inverse FFT is as follows:-

it will be an array mostly filled with zeros

→ Inverse FFT:- It is the exact opposite of FFT. It takes a frequency domain representation and mathematically reconstructs the original image signal back into spatial or time domain.
Ex - If we take FFT of an image and we forcefully change all high frequency values to 0. When we apply inverse FFT the resulting image will be original image but not completely blurring.

Q3) Difference b/w LPF & HPF w.r.t noise & edge

LPF (Low pass)	HPF (High pass)
(1) Allows low frequencies, blocks high frequencies	(1) Allows high frequencies, removes low frequencies
(2) Removes high frequency noise like salt & pepper noise, etc.	(2) Enhances noise because noise is high frequency
(3) It blurs edges because edges are high frequency components.	(3) It enhances edges and makes images sharper.

Q4) Image degradation w.r. to blur & noise means how an image gets distorted during capture, transmission or processing.

Degradation Model

$$g(x, y) = h(x, y) * f(x, y) + n(x, y)$$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
Degraded image Degradation function Original image Noise

$*$ $\rightarrow$ Convolution function